Syllabus 2026-09-09

Macroeconomic Theory I: Intertemporal Choice

Professor Christopher D. Carroll

Office Hours: Make An Appointment Yourself (read the “rules” link)

Teaching Assistant:
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Required discussion section: Thursdays, 1:30–2:20 pm, Wyman W603.
Office Hours: TBA

Link: Link to paperpile folder which contains all the readings for this course.

If you're a Google Calendar user, you should be able to subscribe to the course calendar using the following Calendar ID (after logging into your Google account, copy and paste the text into the “Other Calendars” box in `http://calendar.google.com`):

`jhuecon.org_v2kso2hp1memk0vka1gomvorm8@group.calendar.google.com`

If you are not yet a Google Calendar user, you should become one; items that I post on the calendar will be viewed as having been distributed to the class. If you insist on boycotting Google, however, you can see the calendar if you click **HERE**

Where everything lives. Anything posted on the course calendar counts as distributed to the class; the calendar can also be followed from Outlook or Apple Calendar via this iCal address, or simply viewed.

Lecture notes the book `jhu-econ.github.io/intertemporal-choice`; each section of this syllabus links to its part.

This year's problem sets, exams, and class recordings the private repository `github.com/ccarrollATjhuecon/Choice-2026`. Access is by invitation: send your GitHub username to the TA. Create the account with your JHU address and apply for GitHub Education (free).

Past exams, with answers `github.com/ccarrollATjhuecon/Choice-2026`, in `Exams/` — a rolling window of the six most recent years, each exam both as it was given and with an answer key.

Recordings and summaries every class is recorded; recordings are in Panopto (sign in with your JHED; reach them through Canvas → *Panopto Video*, or the links in the repository's `Classes/README.md`), and the Zoom AI summary and transcript of each class are in the repository's `Classes/` folder.

Computation the Econ-ARK toolkit (HARK) and its demonstration notebooks (DemARK), which also run in the browser with nothing installed at `econ-ark.org/materials`. Bring a laptop to every class. The TA runs a setup session in the first two weeks; installation instructions are distributed before it.

Submitting problem sets as stated in each problem set's README in the repository.

Who to ask anything about the running of the course — repository access, submissions, the section, software, absences — goes to the TA, who is copied on everything; the economics comes to me.

Grading final exam 50%, midterm 25%, problem sets 10%, participation 15%.

Theory and evidence about the dynamic behavior of households, firms, and the macroeconomy as a whole in the short and long run. Begins with a thorough discussion of the consumption/saving problem of households, including the role of uncertainty, then moves to investment behavior of firms, including the relationship between financial markets and firm behavior. General equilibrium models of firms and households combine to generate benchmark models of economic growth, which leads us to a benchmark specification for dynamic aggregate macroeconomic models.

This syllabus contains required and recommended readings for each topic. Required readings are indicated by a $*$.

You should own a copy of the texts by Blanchard and Fischer (1989), Deaton (1992), Romer (2011) and Sargent and Ljunqvist (2012).

Students from other departments are welcome to take the course, but they should be prepared for regular use of higher mathematics (multivariate calculus, probability and statistics, functional analysis) which will be employed later in the course.

Course handouts are available at:

`jhu-econ.github.io/intertemporal-choice`

1 Course Intro: Methodology for Macroeconomists

Bib: Method.bib

Handouts: `Methodology`

Readings:

* Summers (1991)
* Acemoglu (2009)
* Krugman (2012)
* Caballero (2010)
* Barbera (2010)
* Romer (2016)

2 Consumption

Bib: `Consumption.bib`

Handouts: `Consumption`

Readings:

2.1 Introduction and Overview

* Friedman (1957), Chapters I, II, III
* Deaton (1992), Preface; pp. 76-81

2.2 The Life Cycle and Overlapping Generations Models

* Deaton (1992), pp. 1-6
* Blanchard and Fischer (1989), Chapter 3: Introduction, pp. 91-113

  Diamond (1965)

  Feldstein (1974)

* Modigliani (1986)

  Summers (1981)

  Pemberton (1997)

* Carroll and Summers (1991)

2.3 The Infinite Horizon Representative Agent Model

* Deaton (1992), Chapter 3
* Hall (1978)
  Flavin (1981)
  Campbell and Deaton (1989)
* Campbell and Mankiw (1989)
  Hall (1988)
  Carroll, Fuhrer, and Wilcox (1994)

2.4 Consumption with Uncertainty and/or Liquidity Constraints

* Deaton (1992), Chapter 6
  Zeldes (1989)
  Caballero (1990)
  Deaton (1991)
  Toche (2005)
* Carroll (2001b)
* Carroll and Kimball (2007)
* Carroll and Summers (1991)
  Carroll (2023)
  Carroll (2001a)

2.5 Consumption Based Asset Pricing

* Lucas (1978)
* Mehra and Prescott (1985)
  Blanchard (1989)
  Carroll (2008)

3 Investment

Bib: `Investment.bib`

Handouts: `Investment`

Readings:

- * Blanchard and Fischer (1989), Chapter 2, pages 58–69; Chapter 6, pp 291–302
- Hall and Jorgenson (1967)
- * Hayashi (1982)
- * Eberly and Abel (1994)
- * Cummins, Hassett, and Hubbard (1994)

4 Growth

Bib: `Growth.bib`

Handouts: `Growth`

- * Blanchard and Fischer (1989), Chapter 2, pp 37-57
- * Deaton (2013)
- * Sala-i Martin (1990)
- King and Rebelo (1993)
- Phelps (1961)

5 Real Business Cycle Models

Bib: `DSGEModels.bib`

Handouts: `DSGEModels`

- * Blanchard and Fischer (1989), Chapter 1
- * Blanchard and Fischer (1989), Chapter 7
- * Prescott (1986)
- * Summers (1986)

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Readings (click to download .bib file)

Acemoglu, Daron (2009): "The Crisis of 2008: Structural Lessons For and From Economics," *Manuscript, MIT*, See also the discussion at http://economistsview.typepad.com/economistsview/2009/01/acemoglu-the-mo.html.

Barbera, Robert J. (2010): "If It Were a Fight, They Would Have Stopped It in December of 2008," *The Economists' Voice*, 7(2), 3, http://EconPapers.repec.org/RePEc:bpj:evoice:v:7:y:2010:i:2:n:3.

Blanchard, Olivier, and Stanley Fischer (1989): *Lectures on Macroeconomics*. MIT Press.

Blanchard, Olivier J. (1989): "Speculative Bubbles, Crashes, and Rational Expectations," *Economics Letters*, 3, 387–389, http://ideas.repec.org/a/eee/ecolet/v3y1979i4p387-389.html.

Caballero, Ricardo J. (1990): "Consumption Puzzles and Precautionary Savings," *Journal of Monetary Economics*, 25, 113–136, http://ideas.repec.org/p/clu/wpaper/1988_05.html.

——— (2010): "Macroeconomics after the Crisis: Time to Deal with the Pretense-of-Knowledge Syndrome," *Journal of Economic Perspectives*, 24(4), 85–102, https://pubs.aeaweb.org/doi/pdfplus/10.1257/jep.24.4.85.

Campbell, John, and Angus Deaton (1989): "Why is Consumption So Smooth?," *The Review of Economic Studies*, 56(3), 357–373, http://www.jstor.org/stable/2297552.

Campbell, John Y., and N. Gregory Mankiw (1989): "Consumption, Income, and Interest Rates: Reinterpreting the Time-Series Evidence," in *NBER Macroeconomics Annual, 1989*, ed. by Olivier J. Blanchard, and Stanley Fischer, pp. 185–216. MIT Press, Cambridge, MA, http://www.nber.org/papers/w2924.pdf.

Carroll, Christopher D. (2001a): "Death to the Log-Linearized Consumption Euler Equation! (And Very Poor Health to the Second-Order Approximation)," *Advances in Macroeconomics*, 1(1), Article 6.

——— (2001b): "A Theory of the Consumption Function, With and Without Liquidity Constraints (Expanded Version)," *NBER Working Paper Number W8387*, https://www.econ2.jhu.edu/people/ccarroll/ATheoryv3NBER.pdf.

——— (2008): "Recent Stock Declines: Panic or the Purge of 'Irrational Exuberance'?," *The Economists' Voice*, 5, https://www.econ2.jhu.edu/people/ccarroll/opinion/CampbellShillerReduxFinal.pdf.

——— (2023): "Theoretical Foundations of Buffer Stock Saving," *Revise and Resubmit, Quantitative Economics.*

Carroll, Christopher D., Jeffrey C. Fuhrer, and David W. Wilcox (1994): "Does Consumer Sentiment Forecast Household Spending? If So, Why?," *American Economic Review*, 84(5), 1397–1408.

Carroll, Christopher D., and Miles S. Kimball (2007): "Precautionary Saving and Precautionary Wealth," *Palgrave Dictionary of Economics and Finance, 2nd Ed.*, https://www.econ2.jhu.edu/people/ccarroll/papers/PalgravePrecautionary.pdf.

Carroll, Christopher D., and Lawrence H. Summers (1991): "Consumption Growth Parallels Income Growth: Some New Evidence," in *National Saving and Economic Performance*, ed. by B. Douglas Bernheim, and John B. Shoven. Chicago University Press, Chicago, https://www.econ2.jhu.edu/people/ccarroll/papers/CParallelsY.pdf.

Cummins, Jason G., Kevin A. Hassett, and R. Glenn Hubbard (1994): "A Reconsideration of Investment Behavior Using Tax Reforms as Natural Experiments," *Brookings Papers on Economic Activity*, 2, 1–59, Available at http://ideas.repec.org/p/nbr/nberre/1946.html.

Deaton, Angus (2013): "Weak States, Poor Countries," Available at Project Syndicate.

Deaton, Angus S. (1991): "Saving and Liquidity Constraints," *Econometrica*, 59, 1221–1248, https://www.jstor.org/stable/2938366.

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Diamond, Peter A. (1965): "National Debt in a Neoclassical Growth Model," *American Economic Review*, 55, 1126–1150, http://www.jstor.org/stable/1809231.

Eberly, Janice, and Andrew Abel (1994): "A Unified Model of Investment Under Uncertainty," *American Economic Review*, 84(5), 1369–1384, Available at http://ideas.repec.org/a/aea/aecrev/v84y1994i5p1369-84.html.

Feldstein, Martin S. (1974): "Social Security, Induced Retirement, and Aggregate Capital Formation," *Journal of Political Economy*, 82(4), 905–926, Available at http://www.jstor.org/stable/1829174.

Flavin, Marjorie B. (1981): "The Adjustment of Consumption to Changing Expectations About Future Income," *Journal of Political Economy*, 89, 974–1009, http://www.jstor.org/stable/1830816.

Friedman, Milton A. (1957): *A Theory of the Consumption Function*. Princeton University Press.

Hall, Robert E. (1978): "Stochastic Implications of the Life-Cycle/Permanent Income Hypothesis: Theory and Evidence," *Journal of Political Economy*, 96, 971–87, Available at http://www.stanford.edu/~rehall/Stochastic-JPE-Dec-1978.pdf.

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Hall, Robert E., and Dale Jorgenson (1967): "Tax Policy and Investment Behavior," *American Economic Review*, 57, Available at http://www.stanford.edu/~rehall/Tax-Policy-AER-June-1967.pdf.

Hayashi, Fumio (1982): "Tobin's Marginal Q and Average Q: A Neoclassical Interpretation," *Econometrica*, 50(1), 213–224, Available at http://ideas.repec.org/p/nwu/cmsems/457.html.

King, Robert G., and Sergio T. Rebelo (1993): "Transitional Dynamics and Economic Growth in the Neoclassical Model," *American Economic Review*, 83(4), 908–931, Available at http://ideas.repec.org/a/aea/aecrev/v83y1993i4p908-31.html.

Krugman, Paul (2012): "Economics in the Crisis," Commencement Address, University of Lisbon, http://krugman.blogs.nytimes.com/2012/03/05/economics-in-the-crisis/.

Lucas, Robert E. (1978): "Asset Prices in an Exchange Economy," *Econometrica*, 46, 1429–1445, Available at http://www.jstor.org/stable/1913837.

Mehra, Rajnish, and Edward C. Prescott (1985): "The Equity Premium: A Puzzle," *Journal of Monetary Economics*, 15, 145–61.

Modigliani, Franco (1986): "Life Cycle, Individual Thrift, and the Wealth of Nations," *American Economic Review*, 3(76), 297–313, Available at http://www.jstor.org/stable/1813352.

Pemberton, James (1997): "The Empircial Failure of the Life Cycle Model with Perfect Capital Markets," *Oxford Economic Papers*, 49(1), 129–151, Available at http://www.jstor.org/stable/2663734.

Phelps, Edmund S. (1961): "The Golden Rule of Accumulation," *American Economic Review*, pp. 638–642, Available at http://teaching.ust.hk/~econ343/PAPERS/EdmundPhelps-TheGoldenRuleofAccumulation-Afablefo

Prescott, Edward C. (1986): "Theory Ahead of Business Cycle Measurement," *Carnegie-Rochester Conference Series on Public Policy*, 25, 11–44, http://ideas.repec.org/p/fip/fedmsr/102.html.

Romer, David (2011): *Advanced Macroeconomics*. McGraw-Hill/Irwin, fourth edn.

Romer, Paul (2016): "The Trouble With Macroeconomics," Discussion paper, http://paulromer.net, Available at https://paulromer.net/wp-content/uploads/2016/09/WP-Trouble.pdf.

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Toche, Patrick (2005): "A Tractable Model of Precautionary Saving in Continuous Time," *Economics Letters*, 87(2), 267–272.

Zeldes, Stephen P. (1989): "Consumption and Liquidity Constraints: An Empirical Investigation," *The Journal of Political Economy*, 97(2), 275–298, http://userwww.service.emory.edu/~skrause/E511/Zeldes-1989.pdf.